Indian Institute of Technology Gandhinagar

ES116 Principles and Applications of Electrical Engineering

Sem II, 2025-2026, Date: 28/04/2026

End-Semester Examination, Duration: 150 mins, Maximum Points: 60

Instructions:

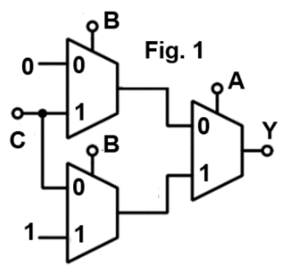
1. This assessment accounts for max. of 20 marks towards your total marks.
2. Show your work clearly. Partial credit may be awarded for correct reasoning.
3. **Stage 1** consists of rapid concept checks. Keep your justifications to 1-2 sentences.
4. **Stage2** consists of analysis questions. Show your logical steps; heavy algebraic derivation is not required (unless specified).

Welcome to the Mars Habitat Project! Your mission is to design the life-critical digital sensor logic and analog oxygen-generation circuits to keep the crew alive.

<u>Stage 1:</u> To qualify to design the digital logic and actuator circuit for the habitat, you must pass this Stage 1 evaluation. Clearly show your work. Keep your justifications short.

Q1 (1 point). An airlock door (D) should only open (D=1) if the Internal Pressure is High (P), the Emergency Lock is OFF (E=0), and *either* a Keycard is present (K=1) or a Manual Override is active high (M). Write the Boolean expression for D.

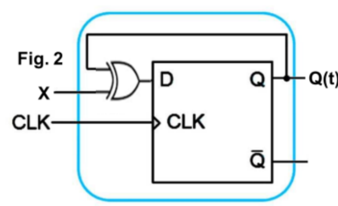
Q2 (2 point). A logic circuit has 3 inputs (A, B, C) where A, B and C denote the binary representation of decimal numbers 0 to 7, A being the most significant bit. The output (Z) is HIGH for decimal values 1, 3, and 5. The decimal values 6 and 7 are "Don't Care" states. Draw the K-Map and find the minimized SOP expression.



Q3 (3 point). You are given a circuit **(Fig. 1)** implemented using 2:1 MUX's. Write the absolute minimum Boolean expression for the output Y in terms of A, B and C. What is the minimum number of inputs that must be HIGH for Y to be 1?

Q4 (2 point). Consider a single D-Flip Flop where the D input is connected to the $\overline{Q}$ output. If the clock (CLK) is a square wave, sketch the Timing Diagram for Q over three clock cycles (assume Q starts at 1). What is the frequency of Q relative to CLK?

Q5 (2 point). A system monitors input (X) with three states: (S_0: IDLE), (S_1: ALERT), and (S_2: ALARM). It moves from S_0 to S_1 when (X = 1), from S_1 to S_2 if X remains 1, and from any state returns to S_0 when X = 0. It holds S_2 while X = 1. Draw the state diagram and determine the minimum number of D flip-flops required.



Q6 (2 point). For the given logic circuit **(Fig. 2),** write the state table where Q(t) represent the Present state, X denotes the Input and, Q(t+1)) represents the Next State. When X=1, what happens to the next state at the clock edge?

Q7 (1 point). A generator requires 0.5 V increments to change speed. If the DAC's reference voltage is 5 V, how many bits' resolution will be sufficient?

Q8 (1 point). For an N-bit binary-weighted resistor DAC, if the resistor connected to the Most Significant Bit (MSB) has a resistance of R, calculate the required resistance for the Least Significant Bit (LSB) in terms of R and N.

Q9 (1 point). Why will an OPAMP with a closed-loop gain of 5 and 20 mA maximum output current fail to give its theoretical 5 V output for an input of 1 V across a 10 Ω motor?

Q10 (1 point). A square wave is applied to an op-amp integrator and the output appears triangular. Explain briefly using frequency response why the output becomes smoother.

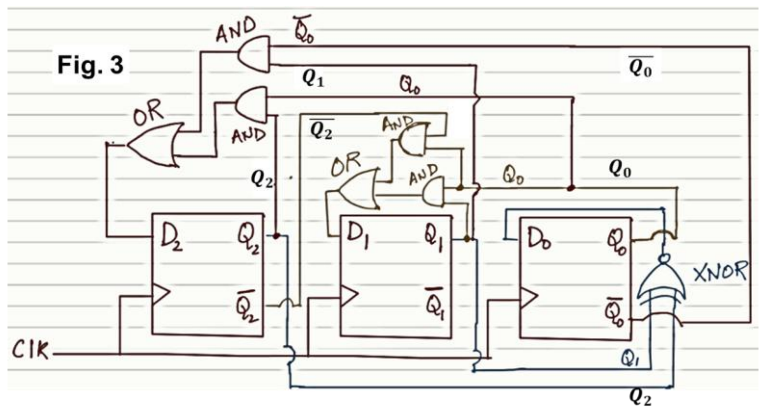
<u>Stage 2:</u>

Q11 (10 points). The habitat has three oxygen sensors (A, B, C). The "Oxygen Low" indicator signal (Y) is active HIGH only if at least two sensors output 1.

A. **(2 point)** Draw the Truth Table.
B. **(2 point)** Minimize using a K-Map.
C. **(3 point)** Implement the circuit using only NAND gates (any number of input).
D. **(3 point)** Implement the circuit using only 2-input NAND gates.

Q12 (14 points). The oxygen generator operates in 8 discrete intensity levels. To ensure smooth transitions between levels and avoid glitches in control signals, the system should give the following sequence 000 → 001 → 011 → 010 → 110 → 111 → 101 → 100 → 000.

A. **(2 points)** Write the State Transition Table.
B. **(6 points)** Derive the D-Flip Flop inputs expressions.
C. **(6 points)** A proposed logic circuit **(Fig. 3)** is intended to implement this sequence. Does it correctly generate the intended sequence? If it fails, identify the specific gate(s) or connections that are incorrect and explain the resulting behavior.



Q13 (6 point). Design a FSM using two D flip-flops with the following state encoding: 00: IDLE, 01: OXYGEN_PROD, 10: VENT. State 11 is unused. State transitions: **1)** From IDLE (00): if Y=1, transition to OXYGEN_PROD (01); otherwise, remain in IDLE, **2)** From OXYGEN_PROD (01): if Z = 1, go to VENT (10); otherwise remain in OXYGEN_PROD, **3)** From VENT (10): return to IDLE unconditionally.

A. **(2 points)** Draw the state diagram.
B. **(4 points)** Draw the state table (D_1, D_0 in terms of Q_1, Q_0, Y and Z). Clearly mark the Don't care terms, if any.

Q14 (10 points). To control the oxygen production, the 3-bit output of Q12 (Q_2, Q_1, Q_0) is used to generate a corresponding analog control voltage using a binary-weighted DAC. Here, Q_2 is the MSB. Logic 0/1 denotes 0 V/5V. Feedback resistor is equal to the MSB resistor.

A. **(4 points)** Design an op-amp based binary-weighted DAC circuit.
B. **(6 points)** Sketch the absolute value of the output voltage ($|V_{analog}|$) for one full sequence given in Q12.

Q15 (4 points). To smooth the DAC output, a capacitor (C_1 = 1 μF) is placed in parallel with the feedback resistor (R_2 = 10 kΩ) of the circuit shown here (R_1 = 5 kΩ).

A. **(2 points)** Calculate the cutoff frequency (f_c) of this filter.
B. **(2 points)** Assume V_{in} shown in the figure which changes much faster than the filter's cutoff frequency ($f_{in} >> f_c$)). Qualitatively sketch the V_{in} and $|V_{out}|$ on the same axes. Justify your answer.